

Arbaz Siddiqui

Principal Software Engineer · AI Infrastructure and Distributed Systems

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SUMMARY

Backend engineer with ten years of experience building consumer products at scale. At Turnip I built and ran the live-streaming and social platform serving **25M users** through four pivots (game streaming, live audio, a creator social network, and now an AI design suite), and spent the last two years on the AI infrastructure around it. Outside work I fine-tune open-weights models, build mobile apps, and use generative AI to run autonomous pipelines, including an open 12B model that trended #1 on Hugging Face and content pipelines at **600K monthly views**.

EXPERIENCE

Turnip · *Principal Software Engineer*

Jan 2021 – Present

Four pivots in one startup, from game streaming to live audio to a creator social network to Zaps, an AI design suite.

- Built and ran the backend serving **25M users**, 20 microservices on Kubernetes exposed through Kong and talking internally over Kafka, with MongoDB as the primary store, Neo4j, Redis and Elasticsearch by use case, and real-time systems over MQTT. Sole backend engineer for the first two years.
- Built the live video streaming at **50K peak concurrent streams** and about **100TB** of video a day on RTMP and FFmpeg at **sub-2s latency**, holding sticky sessions over TCP through HAProxy ingress and packing streams onto shared Kubernetes pods for efficiency and cost.
- Built live audio rooms on Agora with MQTT holding presence and state for **2M monthly users** over **70M call minutes**.
- Built the social network's feed and graph, weighting every interaction by a user and their friends into a per-user ranking cached and served at **sub-100ms**, with the graph on **Neo4j** powering second-degree friend discovery and suggestions.
- Shipped the AI editing suite of face swap, image enhancement and styles on **ComfyUI** with open-weights models, served on in-house GPU inference and then moved to a distributed GPU network, cutting GPU cost **80%** at about **2 seconds** an image and around **100K HD generations a day**.
- Replaced template tag search with semantic retrieval over **40K** embedded templates and carousels, served by vector search at **p95 under 50ms**.
- Built an in-house AI agent harness for internal tooling where each project carries its own memory, TDD requirements, deployment skills and guardrails, a black box to the product team. **Ten non-engineers shipped 17 internal tools** with zero engineer commits.
- Migrated core services from Node to Go behind a strangler fig, with byte-parity responses under an enforced **100% test-coverage** gate.

MouseBelt · *Senior Backend Engineer (Contract)*

Apr 2020 – Jan 2021

- Built a centralized cryptocurrency exchange end to end, accounts, KYC, markets and order flow on a **matching engine written in C** with gRPC and WebSocket layers.
- Ran custody across **six chains** (Bitcoin, Ethereum, EOS, Cosmos, ICON, BNB Chain) and built a multi-chain **DEX aggregator** with price scrapers across BSC, ICON and Moonbeam behind one interface and an on-chain indexer.

India's biggest event ticketing platform.

- Led **seven engineers** across backend and mobile on the movie ticketing platform holding **500+ requests a second** on sale nights, where integrating Vista, the slow and error-prone legacy theatre software, meant redesigning the whole checkout from synchronous to asynchronous.
- Designed the logistics architecture, from internal tooling for the operations team to user-facing order tracking and delivery communication, integrated with logistics partners.
- Built the commerce flow end to end across cart, payments, inventory and taxation, **30 microservices** on EKS with SQS for async communication, Lambda and Kinesis.
- Moved on-demand workloads to serverless on Lambda and Kinesis, cutting cost by over **70%**.

SELECTED PROJECTS

- **Ozan-v1-12B**. Fine-tuned and released a 12B open-weights long-form fiction writing model with QLoRA on one A100 for about \$40. Lowest slop of any 12B on EQ-Bench, beating bigger SOTA models like Gemma 3 27B. **#1 trending on Hugging Face**, 3K+ monthly downloads. hf.co/arbazsiddiqui
- **The Footnote**. An autonomous pipeline that researches, scripts, voices, illustrates, renders and publishes at **600K monthly views**, built entirely on open-weights models and without any human supervision. @thefootnotemedia
- **Iris**. A privacy-first cycle tracker, **4.9★** on the App Store in **twelve languages**, with Apple Health sync and partner sharing. Organic installs come from autonomous women's-health content pipelines on Instagram and Pinterest. [App Store](#) · [Pinterest](#)
- **Mercuro**. A logic puzzle game with its own constraint solver and generator, **660 curated puzzles** on iOS and Android, featured in gaming newsletters.
- **Iru-cache-node**. An O(1) LRU cache with expiry for Node.js, **119★**, featured in Node Weekly. [github](#)
- **anabran**. An HTTP load balancer and reverse proxy written in Go. [github](#)

SKILLS

Languages	Go, TypeScript, JavaScript, Python
AI	RAG, agent orchestration (LangGraph, MCP, Claude Code), fine-tuning (QLoRA), model serving, GPU inference, evals, PyTorch, ONNX, Triton, MLflow, generative image pipelines (ComfyUI), diffusion models (Qwen, MiniMax, Flux), workflow automation (n8n), llama.cpp
Backend	Node.js, Kafka, REST, gRPC, WebSockets, microservices, event-driven architecture, RabbitMQ, SQS, Kinesis, MQTT, FastAPI, RTMP, FFmpeg
Data	MongoDB, PostgreSQL, Redis, Elasticsearch, Neo4j, Cassandra, Firebase RTDB
Infrastructure	Kubernetes (GKE, EKS), Docker, AWS, GCP, Terraform, Helm, Kong, Lambda, CI/CD
Observability	Prometheus, Grafana, ELK, OpenTelemetry, New Relic, Sentry
Crypto	Order matching engines, Multi-chain custody, DEX aggregation, NFT systems

EDUCATION

B.Tech, Computer Science and Engineering · The LNM Institute of Information Technology 2013 – 2017